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Enhancing creative writing through AI-powered co-creation with cognitive and emotional outcomes

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This research explored how generative artificial intelligence (AI) tools, specifically a writing assistant powered by large language models like ChatGPT, influence the development of creative writing (CW) skills in intermediate EFL learners, focusing on both cognitive and emotional aspects. The research included 92 male participants who were divided into three distinct groups: The High-AI Support Group, the Low-AI Support Group, and the Control Group (CG). Participants completed pretesting with some validated scales. After the pretest, the High-AI Support Group was provided with comprehensive AI-generated feedback, while the Low-AI Support Group received limited AI support, and the Control Group adhered to conventional writing instruction. After the treatment, posttests were given to all groups, and eight participants from each experimental group took part in interviews. Quantitative analyses revealed that both groups receiving AI support outperformed the Control Group, with the High-AI Support Group exhibiting the most significant enhancements. Qualitative insights revealed that students in the High-AI Support Group demonstrated increased engagement and creativity, attributed to tailored AI feedback that alleviated cognitive load by addressing technical aspects like grammatical accuracy. In contrast, members of the Low-AI Support Group appreciated the collaborative interaction between AI and instructor feedback, which enhanced their confidence, independence, and skills in grammar and writing conventions. In summary, this study emphasizes the importance of AI in enhancing both intellectual and emotional growth for EFL learners, advocating for its incorporation as a valuable resource in educational settings.

Keywords AI-powered co-creation, Creative writing, Cognitive load, Emotional engagement, EFL learners

The swift progress of artificial intelligence (AI) technologies has significantly impacted multiple areas of education, especially in English as a Foreign Language (EFL) instruction. Tools such as large language models provide creative approaches to improve learning experiences[1]. Recently, generative AI systems like ChatGPT have become valuable tools, offering personalized feedback, generating creative prompts, and enabling interactive language practice. These capabilities can greatly enhance learners' skills in writing, vocabulary acquisition, and grammatical accuracy[2,3]. Creative writing (CW) is essential in EFL education, significantly enhancing linguistic competence by allowing learners to express their emotions, ideas, and stories in imaginative ways. This approach transcends rote memorization, fostering a deeper engagement with the language[4,5]. The incorporation of AI into CW tasks aids cognitive growth by alleviating the mental burden of technical details. Additionally, it fosters emotional engagement (EE), transforming the learning experience into a more enjoyable and motivating journey for students who frequently struggle in conventional classroom environments. While AI has the potential to transform EFL pedagogy, it is crucial to tackle ethical issues like data privacy and bias to guarantee fair advantages for every learner[6,7].

In Iran, where EFL education operates within a distinct cultural and institutional framework, the integration of AI tools offers specific opportunities and challenges that cater to local requirements[1]. Iranian EFL learners, who often find themselves in gender-segregated classrooms and resource-limited settings within urban language institutes, face significant challenges. These include the high cognitive load (CL) associated with grammatical complexities and a lack of authentic English interactions, both of which can impede their creative expression and EE in writing tasks[8]. Recent studies in Iran have shown that integrating AI can address these challenges by providing adaptive, real-time support that aligns with national educational policies focused on technological innovation in language learning. This approach helps bridge the gaps in traditional instruction and fosters learner autonomy, particularly in contexts where access to native speakers is limited. AI-driven platforms have

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been demonstrated to improve writing skills among Iranian EFL students by offering instant corrections and generating ideas, which helps to alleviate anxiety and create a more encouraging learning environment[9,10]. This context highlights AI's ability to adapt to Iran's EFL environment, where cultural sensitivities and teaching traditions shape the perception and use of technology, ultimately supporting the development of skills crucial for global communication.

Despite these advancements, there remains a significant gap in the literature concerning how different levels of AI support affect CW from both cognitive and emotional angles, especially for intermediate EFL learners in non-Western settings such as Iran. Although previous studies have looked into AI's impact on general language abilities, there is a lack of systematic comparisons between high and low AI assistance in CW tasks, particularly regarding their effects on CL management and EE[11,12]. This study fills a gap by examining how generative AI, specifically ChatGPT, influences the CW development of Iranian intermediate EFL learners. It emphasizes the reduction of cognitive demands and enhancement of EE through the use of controlled experimental groups. This study is important because it can guide EFL teaching in Iran and similar settings. It shows how thoughtful integration of AI can enhance learning results, promote ethical use of technology, and support greater educational fairness. The findings present empirical evidence regarding the cognitive and emotional advantages of AI, suggesting practical applications for educators, curriculum developers, and policymakers. By integrating AI as a collaborative resource, it can boost learner motivation and proficiency while maintaining the essential humanistic aspects of education[6,13]. This research enhances our comprehension of AI's significant impact on EFL CW, setting the stage for more inclusive and effective language teaching strategies that resonate with global trends in educational technology. In light of these considerations, the subsequent research questions emerged:

1. Is there a significant difference between the effects of high AI support and low AI support on writing creativity among intermediate EFL learners?
2. Is there a significant difference between the effects of high AI support and low AI support on cognitive load among intermediate EFL learners?
3. Is there a significant difference between the effects of high AI support and low AI support on emotional engagement among intermediate EFL learners?
4. What are EFL learners' perceptions of the effectiveness of low versus high AI support on their creative writing skills, cognitive load, and emotional engagement?

Literature review

Artificial intelligence

The incorporation of AI in language education has attracted considerable interest due to its potential to transform both learning and teaching methods[2,3]. Nonetheless, these advancements bring forth challenges in pedagogy and ethics that necessitate thorough scrutiny. As AI systems gather and analyze extensive learner data, concerns surrounding data privacy, security, and consent gain increased attention[7]. Maintaining transparency in data usage and safeguarding sensitive information is essential for upholding ethical standards[6]. Moreover, it is essential to tackle concerns regarding bias and unfairness, as AI applications may unintentionally perpetuate existing disparities among learners[14,15].

In addition to ethical concerns, the educational impact of AI on language learning requires careful examination. AI-driven programs provide tailored learning experiences, yet their success in improving outcomes necessitates further empirical research[16]. It is essential for researchers to closely examine how students engage with AI platforms, how these tools complement traditional methods, and their impact on boosting participation and motivation[11]. As AI significantly shapes education, it is essential to explore its impact on the personal and professional development of both teachers and learners, as it broadens educational opportunities[13]. In the context of EFL teaching, AI has transformed traditional methods by offering innovative approaches and tools that enhance learning outcomes[17]. The integration of natural language processing (NLP) with AI fosters greater independence, collaboration, and comprehension of linguistic components such as word meanings and sentence structures[1810]. introduced a theoretical framework for integrating AI, merging technology acceptance models with innovation theory, and highlighting the significant relationship between technology and the learning environment. Moreover, the use of AI in EFL enhances language skills, increases learner motivation, and reduces stress, fostering a more supportive and effective learning environment[9].

ChatGPT, a specific form of AI, has garnered significant interest for its capacity to produce conversational and contextually appropriate responses derived from extensive text data[19]. ChatGPT operates as an unsupervised model that enhances the GPT-3 architecture, providing users with dynamic and personalized interactions[20]. This tool mimics human conversation and provides immediate responses, proving to be beneficial in educational settings[21]. ChatGPT is available on various platforms, including websites and apps, allowing for simultaneous interactions with multiple users. This adaptability makes it suitable for both educational and professional settings[22]. The conversational aspect is especially advantageous for language learners, offering tailored support and engaging experiences[23,24]. ChatGPT, as an advanced chatbot, presents notable benefits compared to traditional rule-based systems. Unlike earlier chatbots that were confined to predefined responses, it utilizes NLP, machine learning, and deep learning to enhance its capabilities over time and interact intelligently with users[25]. ChatGPT serves as a valuable resource for second language (L2) learners by offering diverse linguistic input, opportunities for conversational practice, and immediate feedback, which fosters engagement and promotes ongoing learning[22,26]. This degree of interactivity enables learners to enhance their language abilities, broaden their vocabulary, and engage in practice within a lively, authentic setting[23]. Additionally, the chatbot fosters a supportive environment that helps learners overcome shyness, enabling them to practice language skills without the anxiety of being judged[24]. Studies indicate that these advanced chatbots greatly improve language skills and boost learner motivation, positioning them as essential tools in contemporary language education[24]. The

expansion of AI necessitates a careful balance between its applications and their effects on students, teachers, and the educational landscape. This approach allows us to leverage AI's potential for enhancing language teaching while upholding ethical standards and promoting positive results.

Creative writing

CW is an activity driven by aesthetic motivation, characterized by a high level of discipline and personal expression, prioritizing imaginative representation rather than strict adherence to factual accuracy[275]. emphasized that CW prioritizes the expression of emotions, events, characters, and experiences over merely presenting facts, showcasing a unique personal creativity. The focus on individuality and creativity positions CW as an essential resource in language education. CW fosters language development while encouraging a profound engagement with the language, resulting in enhanced skills in grammar, vocabulary, and discourse. Students actively shape language to express their individual thoughts, which leads to a deeper cognitive engagement with linguistic content, resulting in improved grammatical precision and more inventive vocabulary selections[4]. Recent studies have shown that CW fosters creative thinking and emotional expression in EFL learners, leading to lasting motivation and improved communication skills across various educational contexts[28].

Integrating CW into language education presents a variety of benefits[29]. suggests that CW fosters a sense of playfulness, motivating learners to engage with language in creative ways[30]. This engaging method allows students to uncover new insights about both the language and their own identities. Furthermore, CW in schools greatly enhances students' aesthetic reading skills and refines their capacity for analyzing textual construction, incorporating their own writing in the process[31]. suggests that a creative student possesses the ability to think divergently, allowing them to generate innovative ideas and utilize unconventional imagery to convey distinct perspectives. In this context, poetry as a form of CW plays a crucial role in fostering students' creativity. By engaging with poetic language, students explore creative avenues for self-expression, which can enhance their capabilities in various contexts of CW[32]. Moreover, CW significantly boosts various aspects of cognitive and language development. Language learners at any proficiency level in grammar, vocabulary, phonology, and discourse can gain from being encouraged to engage with language in creative and dynamic ways[33]. CW instruction not only enhances writing skills but also fosters creativity in everyday life[3435]. noted that a child's inherent creativity can be either fostered or stifled by the educational environment. Through CW, a student enhances both their understanding and communication skills by exercising autonomy in their writing choices. The flexibility in CW provides an enjoyable and engaging experience for learners, whether in their first or L2[36]. Recent studies highlight that the combination of collaborative writing with contemporary tools like AI enhances these advantages by offering tailored feedback and encouraging independence in EFL writing activities, which in turn increases confidence and diminishes anxiety in creative endeavors[37].

Emotional engagement

Engagement is acknowledged as a crucial component for grasping how students learn in educational environments[38,39]. Students who are highly engaged often perform better academically, gain more knowledge, and are less likely to drop out of their educational pursuits[40–42]. Engagement is often viewed by scholars as a complex and context-dependent concept, encompassing various subtypes including cognitive, behavioral, and EE, which highlight different aspects of students' participation in the learning process[43,44]. In this context, EE holds considerable importance. This includes the positive emotional reactions of learners (i.e., excitement, enthusiasm, enjoyment, and satisfaction) toward instructional environments, educators, and fellow students throughout the learning experience[45]. The affective dimension plays a vital role in learner motivation, showcasing a positive attitude towards both the subject matter and the language[46]. Conversely, emotional disengagement is characterized by negative emotions such as boredom, frustration, and sadness[45]. Recent studies have shed light on the interplay between EE and foreign language emotions, highlighting their impact on learning outcomes, particularly in technology-enhanced educational settings. Research in AI-mediated EFL settings emphasizes the importance of personalized feedback in promoting positive emotional states[47,48]. This highlights the necessity of integrating emotional considerations with cognitive and behavioral aspects to gain a comprehensive understanding of student engagement.

Although it is widely acknowledged as significant, EE has often received less attention than behavioral and cognitive aspects[49]. Its essential contribution to enhancing intrinsic motivation and advancing language learning outcomes is gaining recognition[50]. Engagement is typically understood as involving three interconnected components: behavioral (action), cognitive (thinking), and affective (feeling) dimensions. Together, these elements influence how learners tackle tasks and maintain their efforts[51–53]. EE has increasingly been recognized for its role in creating a positive learning atmosphere, where students feel both motivated and supported[51]. Studies in language learning contexts indicate that learners who are emotionally engaged tend to find more enjoyment, interest, and satisfaction in their tasks[51]. These learners often demonstrate greater purpose, willingness, and autonomy in their language use[46]. Numerous studies have explored the elements that affect EE, such as how learners respond to written corrective feedback[54], the impact of task familiarity and repetition[55] and various factors at the learner, lesson, and task levels across different classroom and online environments[56]. Nonetheless, certain instructional models, like online flipped classrooms, have been noted to enhance behavioral and cognitive engagement, although their impact on EE appears to be more restricted[57]. The ongoing gap highlights the need for focused strategies that validate and nurture students' emotional involvement, fostering more inspiring learning environments[49,58]. Recent studies emphasize the promise of AI-mediated tools in fulfilling this need by offering personalized interactions that alleviate anxiety and enhance enjoyment in EFL writing tasks. Findings suggest that such tools can bolster learner resilience through emotion-aware feedback and adaptive systems[47,59]. The recent advancements indicate encouraging possibilities for incorporating technology that fosters ongoing learner motivation via designs that respond to emotional needs.

Cognitive load

Cognitive Load Theory (CLT) suggests that working memory has a finite capacity, and when instructional demands surpass this threshold, learning efficiency diminishes[60,61]. CL pertains to the mental effort needed to process information and engage in learning tasks within working memory[62]. Therefore, effectively managing these cognitive resources through thoughtful instructional design is crucial. CLT offers a useful lens for exploring the impact of AI interventions on mental effort and performance. As AI technologies are more frequently utilized to enhance instruction, alleviate unnecessary CL, and promote deeper learning[63,64], this framework becomes increasingly relevant. Integrating CLT principles with AI-driven systems can foster efficient, adaptive, and personalized learning environments.

In this context, CLT identifies three main categories of load: intrinsic, which arises from the complexity of the task; extraneous, resulting from inadequate instructional design; and germane, which refers to the constructive cognitive effort aimed at building schemas[65,66]. Effective teaching seeks to reduce unnecessary CL while promoting relevant processing, thereby optimizing the use of limited working memory[67–69]. There is growing evidence that AI tools, including those that offer adaptive scaffolding, automated feedback, and personalized pacing, can effectively manage these types of CL[70]. AI has the ability to tailor task difficulty to align with a learner's proficiency, thereby reducing frustration[71]. Additionally, it can boost germane load by encouraging reflection and supporting schema development. Interactive technologies, such as intelligent tutoring systems, enhance critical engagement and facilitate knowledge building[72]. Educators need to be aware of various challenges, including the impact of learner demographics and technological design on load management[73], as well as ethical issues like data privacy and algorithmic bias[74]. Recent studies highlight promising AI outcomes for learning efficiency[75,76], yet many fail to address its complex impacts on the three dimensions of collaborative learning in English-language education. This study explores how different levels of AI support affect intrinsic, extraneous, and germane CL in EFL learners engaged in CW tasks.

Method and measures

Sample

This study began with 142 EFL learners, from which 92 were chosen as intermediate-level participants based on their performance on the Oxford Quick Placement Test (OQPT). Their scores fell between 30 and 47 out of 60, aligning with the test's criteria for confirming intermediate proficiency. Participants were required to be male EFL learners aged 15 to 20 years, currently enrolled in English language courses at the chosen institutes, and exhibiting intermediate English proficiency as measured by the OQPT. Those excluded included learners scoring below the intermediate level, individuals who did not consent to participate, and those with significant prior exposure to AI writing tools to prevent confounding variables. A total of 92 male learners, ranging in age from 15 to 20 years, were chosen for this study. In Iran, the gender-segregated nature of educational settings meant that we were only able to recruit male participants. After selection, participants were randomly assigned to three groups through a computer-generated randomization process, which aimed to ensure a balanced distribution and reduce selection bias. High-AI Support Group (Experimental Group 1; $N = 31$): Participants received comprehensive AI-generated feedback and support throughout their learning journey. Low-AI Support Group (Experimental Group 2; $N = 31$): Participants received restricted AI-generated feedback with minimal integration of technology. Control Group (CG; $N = 30$): Engaged in conventional teaching methods, lacking any AI-driven feedback or assistance. The convenience sampling method allowed for the recruitment of participants from two well-known English language institutes in Ahvaz, Iran. These institutes were selected due to their accessibility to the authors, a significant number of intermediate EFL learners, and their representation of typical urban language learning environments in the area. This approach supports the feasibility of the study, while also recognizing limitations in terms of broader generalizability[8].

Before taking part, all participants were provided with comprehensive information regarding the study's goals, methods, and their rights. Participation was voluntary, and we ensured confidentiality by anonymizing all data and safeguarding participants' identities. The study not only followed essential ethical guidelines but also took steps to protect adolescent participants by securing assent from minors in conjunction with parental consent and ensuring cultural sensitivity in data management. Informed consent was obtained from all participants before data collection commenced. For participants younger than 18, we also secured written informed consent from their legal guardians.

Measures

The OQPT was the initial instrument given to all participants in this study. The OQPT is a recognized and standardized test designed to evaluate the language proficiency of English learners, focusing on their reading comprehension, vocabulary, and grammar skills. Created by Oxford University Press, the OQPT is an adaptive test that utilizes a computer-adaptive format. This approach modifies the difficulty of questions based on the test-taker's responses, allowing for accurate placement across various proficiency levels in accordance with the Common European Framework of Reference for Languages (CEFR). The test consists of 60 items, with questions thoughtfully allocated across various skill areas to assess overall language proficiency. Research has demonstrated the instrument's strong psychometric properties, highlighted by high internal consistency reliability, with Cronbach's alpha coefficients between 0.90 and 0.92 across various samples, and a solid test-retest reliability of 0.87 over short intervals. Additionally, evidence of validity backs its application, with concurrent validity shown through significant correlations ($r = 0.78$ to 0.85) with recognized proficiency assessments like the TOEFL iBT and IELTS, and construct validity confirmed through alignment with CEFR standards in international validation studies. The primary purpose of the OQPT was to standardize the participants' English proficiency, ensuring that everyone involved in the experiment was approximately at an intermediate level. The test facilitated an effective

grouping of learners, ensuring that the participants selected for the study had comparable proficiency levels. This approach minimized potential variations in language ability that could have influenced the experiment's results.

The Emotional Engagement Scale (EES) was modified from the validated EE survey created by[77] (See Supplementary File A). This scale, comprising eight items, was created to investigate the EE of language learners. The statements included expressions like, "I enjoyed learning new things in the language class" and "I feel good in the language classroom." The scale serves as a comprehensive tool that captures emotional responses such as enjoyment, interest, and satisfaction in language learning environments. It builds on well-established engagement frameworks and has been tailored for EFL contexts to maintain cultural and contextual relevance. To ensure clear understanding among the Iranian participants, the scale was translated into Persian by two translation experts. They adhered to a strict back-translation process to preserve the original meaning of the English version. The scale's content validity for this study was confirmed through a review by a panel of experts specializing in language education and psychometrics. The experts affirmed that the scale effectively encompasses the essential elements of EE pertinent to language learning. Further validity evidence encompasses construct validity, demonstrated by confirmatory factor analysis that indicates a single-factor structure with factor loadings exceeding 0.70. Additionally, convergent validity is established through moderate to strong correlations (r = 0.62 to 0.75) with related constructs, including foreign language enjoyment and motivation scales from previous empirical studies. The scale's reliability was evaluated through Cronbach's alpha, resulting in a high reliability coefficient ($\alpha = 0.92$, $N = 8$). This suggests that the scale consistently measures emotional exhaustion across the items. The strong reliability of this scale indicates that it is a trustworthy instrument for evaluating EE in language learners, thereby enhancing its validity in this research.

The third tool utilized was the Creativity Assessment Scale (CAS), presented in a Likert format, which was adapted from established creativity rubrics[78,79] (See Supplementary File B). This tool and scoring guide assess the originality, coherence, and other vital storytelling elements that characterize important aspects of creativity in storytelling. The scale consisted of 12 items, utilizing a five-point Likert scale that ranged from Strongly Disagree to Strongly Agree. Rooted in established theories of creativity, this scale assesses key elements like fluency, flexibility, elaboration, and aesthetic appeal in written narratives, making it an effective tool for evaluating EFL CW outcomes in educational research. To guarantee that participants comprehended the items within their native linguistic context, the scale was translated into Persian by two translation experts, who included a back-translation step to confirm accuracy to the original material. Expert reviews were conducted to assess content validity, confirming that the tool is suitable for measuring creativity in narrative writing. The instrument's construct validity has been supported by multiple studies, with exploratory factor analysis revealing a multi-factor structure that corresponds with creative domains, and discriminant validity effectively differentiating it from general writing proficiency measures ($r < 0.50$). The scale demonstrates impressive reliability, with a Cronbach's alpha of $\alpha = 0.89$, indicating strong internal consistency. Furthermore, inter-rater reliability coefficients, including an intraclass correlation of 0.82, have been documented in studies assessing expert judgments of creative products, thereby bolstering its credibility for subjective evaluations of creativity.

The fourth instrument, the CLQ, was adapted from the measures developed by[80,81] (See Supplementary File C). The questionnaire consists of 8 items and utilizes a six-point Likert rating scale. It includes 5 items that evaluate "mental load" and 3 items that measure "mental effort." This multi-item adaptation enhances CLT by distinguishing between intrinsic load (task complexity), extraneous load (instructional design), and germane load (learning investment). This differentiation enables a more nuanced measurement in complex tasks such as CW. Considering the participants' EFL background, two translation experts translated the questionnaire items into Persian, and back-translation was used to ensure semantic and conceptual accuracy between the languages. The questionnaire's validity was established via content validity, involving an expert review to confirm that the items accurately represent the constructs of mental load and mental effort. Factor analysis further affirmed construct validity, demonstrating that the two dimensions were indeed separate. Comprehensive validation studies have confirmed its criterion validity, showing correlations (r = 0.55 to 0.70) with objective physiological measures such as heart rate variability and pupil dilation in multimedia learning contexts, along with its ability to predict learning performance outcomes. The internal consistencies of the two dimensions were robust, with Cronbach's alpha recorded at 0.86 for "mental load" and 0.85 for "mental effort." The data suggest that the questionnaire is a dependable tool for assessing CL within the study's context.

The lead researcher conducted the semi-structured interviews for the qualitative section. For these interviews, eight participants were randomly chosen from each experimental group: the High-AI Support Group and the Low-AI Support Group (Table 1). The sample size adhered to established guidelines, indicating that six to ten participants are practical for qualitative inquiries in classroom settings[82,83]. Each interview, conducted in English, spanned a duration of 19 to 33 min and was held face-to-face. The interview protocol utilized a structured yet flexible format, developed collaboratively by the research team based on initial quantitative findings and themes from AI-related EFL literature. It was tested with non-study EFL learners to enhance question clarity and cultural relevance, adhering to established practices in qualitative research within applied linguistics. The inquiries focused on the experiences of participants in AI-enhanced versus conventional English instruction, highlighting the impact of AI on CW, emotional involvement, and cognitive demands. To ensure clarity, some important sample questions were as follows: (1) In what ways did the level of AI support impact your CW process? Please provide specific examples. (2) How did your interactions with AI influence your emotional responses, such as feelings of excitement or frustration, while completing tasks? (3) In what ways has AI influenced your CL, particularly regarding grammar management, allowing you to concentrate more on your ideas? What challenges have you encountered in this process? (4) What benefits or limitations did AI offer in comparison to conventional approaches for developing writing skills, autonomy, and confidence? (5) What suggestions do you have for enhancing AI integration in EFL CW to better assist learners with similar needs? The semi-structured design preserved essential questions for consistency, while also permitting exploration of emerging themes,

Participant	Gender	Age	Years of learning english	Group
1	Male	16	4	High-AI Support Group
2	Male	17	5	High-AI Support Group
3	Male	18	6	High-AI Support Group
4	Male	16	5	High-AI Support Group
5	Male	19	7	High-AI Support Group
6	Male	20	8	High-AI Support Group
7	Male	18	6	High-AI Support Group
8	Male	17	5	High-AI Support Group
9	Male	16	4	Low-AI Support Group
10	Male	17	5	Low-AI Support Group
11	Male	15	6	Low-AI Support Group
12	Male	16	5	Low-AI Support Group
13	Male	19	7	Low-AI Support Group
14	Male	20	8	Low-AI Support Group
15	Male	18	6	Low-AI Support Group
16	Male	17	5	Low-AI Support Group

Table 1. Demographic information of participants for semi-structured interviews.

which led to richer insights into emotional reactions to AI co-creation in writing. We maintained consistency and reliability by using uniform introductory scripts that emphasized purpose and confidentiality, along with neutral probing in quiet, institute-based settings.

Data collection procedure

The data collection process began with the selection and random assignment of 92 male EFL learners into three groups: the High-AI Support Group ($n = 31$), the Low-AI Support Group ($n = 31$), and the Control Group ($n = 30$). To ensure baseline equivalence, all participants undertook three pretests to evaluate their initial CW skills, CL, and EE. These assessments were conducted using the CAS, CLQ, and EES, respectively. The treatment phase comprised one 90-minute session for each group, which included a 30-minute training period and a subsequent 60-minute CW task. This structure was intended to provide controlled exposure to the interventions while facilitating meaningful engagement.

Members of the High-AI Support Group made extensive use of ChatGPT, a generative AI tool developed by OpenAI based on the GPT architecture, as a writing assistant in their CW tasks. Prior to the main task, participants engaged in a 30-minute training session. During this time, they were introduced to interacting with ChatGPT through guided demonstrations that focused on generating story prompts, refining sentences, and obtaining real-time feedback on grammar and style. For example, participants could ask ChatGPT for imaginative plot suggestions, such as "Write a story about an unexpected friendship between a robot and a human," or to refine sentences for improved tone or complexity, like changing "He was very angry" to "He clenched his fists, his face turning crimson with suppressed rage." Throughout the 60-minute writing task, participants utilized ChatGPT to generate ideas, complete sentences, and assess their drafts for clarity and coherence. A participant could initiate a story with an opening line and then seek a continuation from ChatGPT, enabling them to concentrate on crafting the narrative while leveraging AI-generated expansions. To uphold ethical standards and protect data privacy, interactions were carried out through individual accounts, avoiding the sharing of personal information, and prompts were restricted to writing-related inquiries. Upon finishing the task, participants took a moment to reflect on their experience with ChatGPT by writing a short journal entry. They concentrated on how it contributed to their creativity and eased CL, providing additional qualitative insights to guide interview selections.

In the Low-AI Support Group, participants utilized the ChatGPT tool with limited guidance and interaction, ensuring consistency in the AI platform while adjusting the level of support provided. Participants engaged in a 30-minute training session, during which they explored fundamental functions, including how to request broad topic suggestions from ChatGPT and generate a list of keywords pertinent to their narrative. For instance, participants could have been asked, "What are some themes for a story about resilience?" During the 60-minute writing task, participants were instructed to use ChatGPT sparingly, limiting its role to initial brainstorming or brief prompts without relying on it for sentence generation or in-depth feedback. A participant was asked to create a story theme, but they had to independently develop the plot, characters, and language. This approach highlighted the importance of participants tapping into their own creativity while utilizing ChatGPT as an additional resource. Guidelines were established to prevent excessive reliance, including a limit of three prompts per task. Upon finishing the task, participants considered how the restricted support from ChatGPT affected their creative journey, especially regarding motivation and problem-solving, documenting their thoughts through journal entries to record their immediate impressions.

The Control Group participants refrained from using any AI throughout the study, providing a baseline for comparison. They participated in a 30-minute training session focused on general CW strategies, including brainstorming techniques, story structure, and the use of sensory details to improve narrative quality. For

example, participants explored the art of creating compelling characters and crafting engaging plot twists through guided examples and collaborative discussions. In the course of the 60-minute writing task, participants crafted a short story on their own, free from any external help. The members of this group drew solely on their own insights, mental effort, and writing abilities to accomplish the task. Following the task, participants engaged in reflective journaling about their creative processes, the challenges they faced, and their overall satisfaction, aligning with the reflection step observed in the experimental groups to maintain procedural consistency.

After the treatment, all participants completed three post-tests that mirrored the pre-tests in structure and content, aimed at assessing changes in CW skills, CL, and EE. Following this, 16 participants (i.e., eight chosen at random from the High-AI Support Group and eight from the Low-AI Support Group) took part in semi-structured interviews to delve deeper into their experiences. The interviews, ranging from 19 to 33 min, were conducted in person in English by the lead researcher in a quiet institute environment. They were audio-recorded with consent, transcribed verbatim into English, and analyzed thematically to uncover recurring patterns and insights.

Data analysis

Quantitative analysis

The team employed SPSS Statistics Version 29.0 for all quantitative analyses. This study utilized descriptive statistics, like means and standard deviations, to highlight data patterns, alongside inferential statistics such as ANOVA and MANOVA to assess group differences and test the research hypotheses. ANOVA was used to compare means among the three groups for each dependent variable, as it effectively identifies significant differences and allows for post-hoc analyses to determine specific group comparisons. MANOVA was utilized to analyze the dependent variables together, taking into account their possible correlations and providing a comprehensive perspective on the effects of the AI intervention. This approach aids in reducing inflated error rates in multi-variable educational research[84]. The assumptions for ANOVA and MANOVA were carefully examined prior to analysis to ensure the appropriateness of these parametric tests. The assumptions for ANOVA were satisfied: observations were independent, thanks to randomized grouping and distinct treatments; the Shapiro-Wilk test indicated normality with $p > 0.05$ for all variables and groups; Levene's test confirmed homogeneity of variances with $p > 0.05$. The assumptions for MANOVA were satisfied: Mahalanobis distances indicated multivariate normality without any outliers; Box's M test showed that the covariance matrices were homogeneous ($p > 0.05$); and the correlations among variables were below 0.70, confirming the absence of multicollinearity.

Qualitative analysis

Thematic analysis was utilized to examine the qualitative data, as outlined by[85], which describes it as a systematic approach for identifying and interpreting patterns. This approach was chosen for its flexibility in examining interview narratives, providing in-depth insights into participants' perspectives on AI in EFL writing without being limited by strict theoretical frameworks. The analysis progressed through six steps, as outlined by[85] framework. The first step included becoming acquainted with the data by reading the transcripts multiple times and taking preliminary notes. In the second step, we generated initial codes by tagging significant data elements, such as "AI prompts new plot twists" to highlight creativity enhancements. In the third step, organize the codes into preliminary themes. For instance, you might group "eased technical burdens" and "freed creative focus" under the theme of load reduction. In step four, we examined the themes to ensure they aligned with the dataset, refining or discarding those that lacked support. In step five, themes were clearly defined and named, such as "interactive AI enhances affective involvement" to address engagement. The sixth step culminated in the report, skillfully intertwining themes with relevant literature and insightful quotes. Examples of coding include: "AI complements teacher guidance," derived from comments on low AI fostering autonomy, which connects to the hybrid support theme; "real-time feedback excites writing," coded for high AI, contributing to the motivation theme. Thematic analysis was employed to examine the recurring themes in participants' views on AI as a creative partner. The interviews underwent transcription and were analyzed systematically. To ensure qualitative reliability, we achieved inter-coder agreement with a Cohen's kappa of 0.85 based on 25% of the transcripts[86] and conducted member checking of theme summaries for participant validation.

Results

Quantitative results

The findings from the ANOVA and MANOVA analyses are presented in the tables below, offering a clear comparison of pretest and posttest results across the dependent variables. Table 2 presents the descriptive

Group	Creative writing mean (SD)	Emotional engagement mean (SD)	Cognitive load mean (SD)
High-AI support group (EG1)	28.22 (4.15)	23.96 (3.82)	26.25 (4.01)
Low-AI support group (EG2)	29.77 (4.28)	24.61 (3.95)	25.58 (3.89)
Control group (CG)	30.80 (4.32)	25.56 (4.10)	25.03 (3.76)

Table 2. Pretest descriptive statistics for creative writing, emotional engagement, and cognitive load across groups. Note. The means and standard deviations (SD) indicate baseline levels, showing that the groups had similar starting points with no significant differences in pretest scores, which supports the validity of random assignment.

Variable	F-value	*p*-value	Partial eta squared
Creative Writing	1.30	0.27	0.03
Emotional Engagement	0.37	0.69	0.01
Cognitive Load	0.71	0.49	0.02

Table 3. Pretest Inferential statistics (ANOVA Results) for creative writing, emotional engagement, and cognitive load. Note. With degrees of freedom (df) set at 2, 89 for between-groups, the absence of significant differences ($p > 0.05$) suggests that the groups were equivalent at baseline.

Group	Creative writing mean (SD)	Emotional engagement mean (SD)	Cognitive load mean (SD)
High-AI Support Group (EG1)	44.00 (5.12)	36.29 (4.85)	14.51 (3.24)
Low-AI Support Group (EG2)	38.74 (4.96)	31.38 (4.62)	19.29 (3.58)
Control Group (CG)	33.70 (4.74)	27.73 (4.39)	23.16 (3.92)

Table 4. Posttest descriptive statistics for creative writing, emotional engagement, and cognitive load across groups. Note. Means and standard deviations (SD) illustrate the effects of the intervention; increased AI support was associated with better outcomes in CW and emotional engagement, as well as a decrease in cognitive load.

statistics for pretest scores in CW, EE, and CL among the three groups, providing a baseline for assessing group equivalence prior to the intervention.

Table 2 illustrates that the control group attained the highest pretest mean score for CW, recording a score of 30.80. This was followed by the low-AI support group with a score of 29.77, and the high-AI support group, which scored 28.22. The pretest scores suggest that the groups began with comparable skills in CW, revealing no significant initial disparities. The control group demonstrated the highest pretest mean for EE at 25.56, followed by the low-AI support group at 24.61, while the high-AI support group was slightly lower at 23.96. The initial similarity in EE among groups suggests that participants started with similar levels of involvement in the CW task. The high-AI support group exhibited the highest pretest mean CL at 26.25, compared to 25.58 for the low-AI support group and 25.03 for the control group, indicating slightly lower levels in the latter two groups. Prior to the intervention, all groups exhibited similar levels of CL during the CW task, establishing a fair basis for evaluating the effects of the treatment.

Table 3 displays the results of ANOVA tests conducted on pretest scores for CW, EE, and CL, indicating that there are no significant baseline differences among the groups.

The ANOVA test conducted on pretest scores for CW showed no significant differences among the groups, with a F value of 1.30 and a p value of 0.27, suggesting that the CW abilities were comparable before the intervention. In a similar vein, the ANOVA analysis for EE produced nonsignificant outcomes, with $F = 0.37$ and $p = 0.69$, indicating that EE levels were comparable among the groups at the outset. No significant difference was found between the groups regarding CL in the pretest, with $F = 0.71$ and $p = 0.49$.

Table 4 presents the descriptive statistics from the posttest, showcasing means and standard deviations for CW, EE, and CL among the three groups, highlighting the changes observed after the intervention.

Table 4 presents the posttest means and standard deviations for CW, EE, and CL among the high-AI support group, the low-AI support group, and the control group. In CW, the high-AI support group achieved a mean score of 44.00, surpassing the low-AI support group at 38.74, and the control group, which scored 33.70. This trend suggests that AI assistance, especially at higher levels, has positively impacted CW abilities. In terms of EE, the high-AI support group achieved a mean score of 36.29, while the low-AI support group scored 31.38, and the control group recorded a mean of 27.73. The findings indicate that AI assistance likely fostered a deeper emotional connection to the writing process, resulting in increased engagement. The high-AI support group exhibited the lowest mean CL at 14.51, indicating a decrease in mental effort attributed to AI assistance. The CL increased steadily to 19.29 for the low-AI support group and 23.16 for the control group, suggesting that AI assistance made the writing task less cognitively taxing.

Table 5 presents the multivariate tests, highlighting the overall impact of group assignment on the combined dependent variables of CW, EE, and CL.

Table 5 presents the multivariate tests that illustrate the overall impact of the independent variable on the combined dependent variables of CW, EE, and CL. The results indicate that the groups variable has a significant effect on the outcome measures, as evidenced by Pillai's Trace of 0.61 and Wilks' Lambda of 0.38, with $p < 0.001$. The substantial effect size, indicated by a partial eta squared of 0.30, demonstrates that the AI intervention had a significant influence on the three measured factors. Furthermore, all tests, including Hotelling's Trace and Roy's Largest Root, demonstrate significance for the groups variable at $p < 0.001$, indicating that the AI intervention had a statistically significant impact on CW, EE, and CL.

Table 6 presents the tests of between-subjects effects, evaluating the distinct influence of group assignment on each dependent variable individually.

Table 6 presents the between-subjects effects, assessing how different groups influence CW, EE, and CL in this analysis. The F-value for groups in CW is 20.00 with a p-value of less than 0.001, indicating a significant

Test	Value	F-value	df	*p*-value	Partial eta squared
Pillai's Trace	0.61	13.45	6, 178	< 0.001	0.30
Wilks' Lambda	0.38	13.45	6, 176	< 0.001	0.30
Hotelling's Trace	1.60	13.45	6, 174	< 0.001	0.30
Roy's Largest Root	1.60	13.45	3, 89	< 0.001	0.30

Table 5. Multivariate tests for group effects on combined dependent variables. Note. All tests show significant overall group effects ($p < 0.001$), with a large effect size, highlighting the considerable impact of the AI intervention.

Variable	F-value	df	*p*-value	Partial eta squared
Creative writing	20.00	2, 89	< 0.001	0.31
Emotional engagement	22.29	2, 89	< 0.001	0.33
Cognitive load	22.77	2, 89	< 0.001	0.33

Table 6. Tests of between-subjects effects for group on creative writing, emotional engagement, and cognitive load. Note. Notable differences ($p < 0.001$) with substantial effect sizes affirm the impact of AI support on each variable.

Variable	Group comparison	Mean difference	*p*-value
Creative writing	EG1 vs. CG	10.30	< 0.001
	EG1 vs. EG2	5.25	< 0.05
	EG2 vs. CG	5.05	< 0.05
Emotional engagement	EG1 vs. CG	8.55	< 0.001
	EG1 vs. EG2	4.90	< 0.05
	EG2 vs. CG	3.65	< 0.05
Cognitive load	EG1 vs. CG	−8.65	< 0.001
	EG1 vs. EG2	−4.78	< 0.05
	EG2 vs. CG	−3.87	< 0.05

Table 7. Posttest multiple comparisons (Scheffé) for creative writing, emotional engagement, and cognitive load. Note. EG1 represents the High-AI Support Group, EG2 denotes the Low-AI Support Group, and CG stands for the Control Group. Positive differences reflect higher scores for the first group, while negative differences indicate lower scores (e.g., reduced cognitive load). All comparisons were significant at $p < 0.05$ or better.

difference in scores across the three groups. A partial eta squared value of 0.31 indicates a large effect size, highlighting the significant impact of the AI intervention on CW. The analysis reveals that EE is significantly impacted by the AI intervention, with an F-value of 22.29 ($p < 0.001$) and a partial eta squared of 0.33. The significant effect size underscores the important role that AI support plays in enhancing EE throughout the task. The analysis of groups related to CL yielded an F-value of 22.77 with a p-value of less than 0.001, indicating significant effects on CL. The partial eta squared value of 0.33 indicates that AI significantly reduced CL for those in the high-AI support group.

Table 7 presents the Scheffé post hoc multiple comparisons for posttest scores, highlighting the specific pairwise differences among groups for each dependent variable.

Table 7 presents the Scheffé posttest multiple comparisons concerning CW, EE, and CL. This analysis explores the average differences among the various groups, specifically the high-AI support group, low-AI support group, and control group, across all dependent variables. In CW, the high-AI support group outperformed both the low-AI support group and the control group, demonstrating significant mean differences of 10.30 when compared to the control group and 5.25 when compared to the low-AI support group. This evidence supports the notion that higher levels of AI assistance led to improved CW. The high-AI support group demonstrated notably higher EE levels than both the low-AI support group and the control group, with mean differences of 8.55 when compared to the control group and 4.90 when compared to the low-AI support group. The findings suggest that AI assistance heightened EE in the writing process. The high-AI support group reported notably lower CL compared to both the low-AI support group and the control group, with a significant mean difference of −8.65 when contrasted with the control group. This finding indicates that AI assistance has reduced the mental effort required for the writing task.

Qualitative results

High-AI support group

Increased EE in writing Members of the High-AI Support Group expressed that they experienced greater emotional involvement in their CW tasks, attributing this to the interactive and responsive qualities of the AI tools. The feedback and suggestions from the AI fostered a greater sense of investment in enhancing their writing skills. The tailored feedback appeared to cultivate a stronger bond with their writing, inspiring them to persist in their efforts.

Example 1 "*I think the AI helps me come up with new ideas. It gives me ideas that make me want to keep writing and make my story better. It seems like I have a writing coach just for me.*"

Example 2 "*The AI feedback makes me feel like someone is really looking at what I've done. It makes me want to write more and do better.*"

Example 3 "*The AI's encouragement helps me keep going when I get stuck. It doesn't just feel like a tool; it feels like a conversation.*"

Enhanced creativity and originality Numerous participants observed that the AI tools ignited their creativity, enabling them to generate original ideas and investigate various narrative avenues. The AI's suggestions inspired them to think creatively and motivated them to craft more original storylines. The AI encouraged students to explore innovative concepts and propose unexpected plot developments, prompting them to move beyond conventional storytelling methods.

Example 1 "*The AI showed me different ways to move the story forward. It made me think of things I never would have thought of on my own, like changing the character's reasons for doing things or adding surprises.*"

Example 2 "*It helped me see my plot in new ways. I was having trouble coming up with a more exciting ending, but the AI gave me an idea that changed the whole story.*"

Example 3 "*The AI told me to make my characters more interesting. Some of the character traits it suggested were things I would never have thought of, but they really made the story better.*"

Perceived CL reduction Participants in the High-AI Support Group shared their experiences of how AI tools alleviated CL by providing immediate feedback. This support enabled them to concentrate more on the creative elements of writing, rather than being preoccupied with grammar or structure. The AI's capacity to tackle technical writing aspects enabled students to focus their efforts on improving narrative and storytelling elements.

Example 1 "*I can focus on making the story better because the AI fixes the spelling and grammar mistakes. It makes writing less stressful and more fun.*"

Example 2 "*I don't have to worry about making little mistakes. The AI fixes them, so I can just think about how to make my story and characters better.*"

Example 3 "*When the AI fixes my grammar, I can focus on making the story more emotional instead of getting stuck on language problems.*"

Low-AI support group

Complementary support with teacher feedback Members of the Low-AI Support Group discovered that the limited AI assistance served as a valuable addition to the feedback provided by their teachers. Although the AI did not take control of the learning process, its involvement proved advantageous in reinforcing essential concepts and offering further clarification. For instance, when a student got feedback from the teacher regarding sentence structure enhancement, the AI provided targeted suggestions on word choice or syntax, thereby reinforcing the teacher's advice. The integration of AI alongside teacher involvement fostered a sense of confidence and support among students in their educational journey. A participant remarked, "*The AI helped me work on things my teacher said I needed to work on, like making my sentences clearer. It didn't seem like the AI was taking the place of the teacher; it seemed more like it was helping me understand what the teacher was saying.*"

Enhanced Understanding through targeted AI assistance Despite the limited assistance provided by AI, participants noted that the feedback effectively directs students to their areas for improvement, especially in writing and grammar. The idea was that AI could assist teachers with suggestions, provide students with a chance to reflect more thoroughly on their work, and facilitate more informed corrections. For instance, when a teacher pointed out grammatical errors in a student's essay, the AI promptly provided feedback concerning verb tense usage that clarified the mistake. A participant shared, " *I had trouble with grammar, but the AI would point out specific mistakes and tell me why they were wrong. It helped me figure out what I needed to work on.*"

Increased confidence and autonomy in learning The members of the Low-AI Support Group valued the AI guidelines for encouraging them to take ownership of their learning, all while benefiting from teachers' feedback. This approach fostered autonomy and self-reliance, allowing students to navigate independently while leveraging insights from both AI and their teachers to enhance their confidence in their abilities. One student

shared his experience of incorporating AI suggestions during the revision of a writing assignment: " *The AI gave me ideas, but I made the final choice about how to change my work. It made me feel like I was in charge of my own learning.*" The combination of AI and teacher involvement encouraged greater independence among learners, allowing students to make their own choices while still benefiting from expert guidance.

Discussion

This research question investigates whether high AI support significantly differs from low AI support in influencing writing creativity among intermediate EFL learners. The results from the quantitative analyses, which included ANOVA and post-hoc Scheffé tests, revealed significant differences among the groups. The High-AI Support Group achieved the highest posttest mean score of 44.00, followed by the Low-AI Support Group at 38.74, and the Control Group at 33.70. This indicates that increased AI support contributed to greater enhancements in CW skills. The findings indicate that significant AI assistance leads to greater improvements in originality, coherence, and narrative aspects through real-time, tailored suggestions. In contrast, limited AI support provides some advantages, though to a lesser degree, by promoting independent creativity in conjunction with human guidance. The findings are consistent with[87], who noted improvements in writing proficiency due to ChatGPT's personalized feedback[88];, who observed enhancements in various aspects of writing through AI mediation[33];, who highlighted advancements in creative storytelling through AI collaboration; and[89], who regarded AI as a catalyst for creativity. This aligns with the findings of[90], who observed advantages in mixed human-AI models for fostering creativity, as well as[91], who emphasized the thorough feedback provided by ChatGPT in promoting originality. The enhanced experimental performance can be attributed to AI's ability to personalize, provide relief, and foster engagement, facilitating a dialogue between learners and AI[87,92]. The findings are supported by the theoretical foundations of CW discussed in the literature review[4,5]. highlight CW as an imaginative activity that benefits from tools for language play and expression, which aligns with the role of AI in sparking ideas. Additionally, CLT[62,65] illustrates how AI reduces extraneous load from technical issues, allowing for more cognitive resources to be devoted to creative schema building, as noted by[63,64]. Furthermore, AI's adaptive support in EFL, as discussed by[2,3], accounts for variations in creativity gains. Vygotsky 's Social Constructivist Theory[93] views AI as a supportive tool within the Zone of Proximal Development[88].

The second research question explored whether there is a significant difference in CL between high AI support and low AI support among intermediate EFL learners. The results revealed notable differences among the groups. The High-AI Support Group had the lowest posttest CL mean at 14.51, followed by the Low-AI Support Group at 19.29, and the Control Group at 23.16. This indicates that greater AI support significantly lessened mental effort during CW tasks. The findings indicate that comprehensive AI feedback reduces cognitive demands by automating grammar and structure corrections. This allows individuals to focus on higher-order thinking. In contrast, limited AI support lessens the load compared to traditional methods, but to a lesser extent, by offering targeted assistance without compromising independence. The findings align well with the work of[94], who demonstrated that AI feedback effectively manages mental effort by automating lower-order tasks. Similarly[60], investigated ChatGPT's role in CL management within academic support. Additionally[67,71], highlighted how AI designs can reduce extraneous load, while[72,73] focused on the importance of adaptive scaffolding in regulating CL. The limited AI assistance enhanced grammar comprehension, aligning with the findings of[90] regarding hybrid models, as well as[91] on the detailed feedback provided by ChatGPT. The findings are supported by CLT as discussed in the literature review[62,65,66]. categorize load types, highlighting how AI can reduce extraneous elements to enhance working memory, a point reinforced by[70]. Furthermore, the role of AI in EFL contributes to stress reduction and the creation of efficient environments, as noted by[9,10]. This suggests that higher AI integration leads to greater load reduction, while ChatGPT's real-time interactions[19,20] help alleviate strain, reflecting differences across groups. CLT, as proposed by[81], helps reduce unnecessary cognitive burden, thereby fostering creative engagement[94].

The third research question examined whether there is a notable difference in EE among intermediate EFL learners when comparing high AI support to low AI support. The analyses yielded noteworthy differences, showing that the High-AI Support Group achieved the highest mean for posttest EE at 36.29, followed by the Low-AI Support Group at 31.38, and the Control Group at 27.73. This suggests that increased AI support enhances enthusiasm, enjoyment, and motivation during the writing process. The findings indicate that extensive interactions with AI foster a supportive and non-judgmental atmosphere, which enhances positive emotional responses and maintains engagement. In contrast, minimal AI support increases involvement beyond conventional teaching methods by supplementing teacher contributions, albeit with a lower level of emotional investment. The findings align with those of[95], who demonstrated that ChatGPT enhances motivation[96];, who connected ChatGPT to self-efficacy and enjoyment[97];, who observed that chatbots reduce anxiety[98];, who underscored the motivational value; and[56,99], who focused on the engaging nature of AI courses. In line with the findings of[98,99], ChatGPT fosters a sense of authenticity and encourages engagement. Further insights from[96] regarding self-efficacy[97], on anxiety reduction[100], concerning critical thinking, and[89] on creative stimulation underscore the role of AI as a complement to human instruction, as supported by[101,102], and[103]. The findings are supported by the foundations of EE discussed in the literature review[51–53]. describe EE as a multidimensional construct, with AI enhancing emotions such as excitement[45,50]. The conversational nature of ChatGPT helps to alleviate shyness and fosters engagement[22,23], which accounts for the higher levels of EE observed in high AI contexts. Additionally, Self-Determination Theory (SDT)[104] demonstrates how AI meets the needs for autonomy and competence, thereby motivating users, as noted by[49,57]. SDT[104] elucidates the dynamics of motivation through the lenses of autonomy and competence. Meanwhile, effective feedback principles[105] highlight the importance of immediacy and specificity.

The last research question examined how EFL learners perceive the effectiveness of low versus high AI support in relation to their CW skills, CL, and EE. The qualitative findings from semi-structured interviews

indicated that participants in the High-AI Support Group found extensive AI to be very effective in enhancing creativity, alleviating CL through immediate feedback, and boosting EE through personalized encouragement. In contrast, members of the Low-AI Support Group appreciated minimal AI for fostering autonomy and confidence by supplementing teacher feedback, although they considered its impact to be less transformative overall. These perceptions suggest that learners value AI for enhancing the enjoyment of writing and reducing its challenges. When AI is viewed as a "personal coach," it garners significant support, while being seen as a "supplementary tool" reflects lower engagement. This dynamic fosters positive attitudes, with more robust enthusiasm stemming from the interactive depth offered by high levels of AI support. The perceptions align with the findings of[87,92], which highlight the motivational effects of AI[106]. demonstrate that combined feedback optimizes writing perspectives[91]. reveal that AI feedback improves both quality and attitudes[107]. examine the emotional promises of chatbots, while[100] emphasizes the democratizing potential of AI. Positive views highlight AI's ability to provide real-time, unbiased support that fosters confidence[107], aligning with[108] regarding motivational benefits. Hybrid feedback leads to improvements in control[91]. In conclusion, improved performance arises from the immediacy, precision, and supportive tone of AI feedback, which fosters creativity, self-efficacy, and engagement[109]. This is supported by research[87,91,96,110] that highlights the transformative impact of human-AI collaboration in EFL writing. The findings are supported by the literature review's exploration of AI ethical and pedagogical implications[6,7], highlighting that personalized interactions foster engagement in an ethical manner. Social Constructivist Theory[93] positions AI as a supportive tool within the Zone of Proximal Development[88], elucidating the effectiveness of advanced AI in fostering creativity and experiential engagement. Additionally, the dimensions of autonomy[111] and engagement[40,46] highlight varying perceptions, where lower levels of AI promote self-regulation while higher levels enhance emotional impact. The Theory of Autonomy[111] elucidates how a low-AI balance can promote self-regulation[98,100].

Conclusions and implications

This study highlights significant advantages of integrating AI-based support, particularly ChatGPT, in enhancing EFL learners' CW skillss and emotional involvement, while also reducing CL. The findings revealed notable differences among the groups. The High-AI Support Group achieved peak posttest means of 44.00 in CW and 36.29 in EE, alongside a minimal CL of 14.51. The Low-AI Support Group recorded means of 38.74, 31.38, and 19.29, while the Control Group had means of 33.70, 27.73, and 23.16. Interviews with the High-AI Support Group highlighted a significant increase in EE due to personalized feedback that ignited motivation. Participants also reported enhanced creativity from innovative prompts and plot suggestions, while the automation of grammar and structure reduced CL, allowing for a greater focus on narrative development. In contrast, participants in the Low-AI Support Group expressed appreciation for minimal AI, noting its ability to complement teacher guidance in fostering autonomy, confidence, and grammatical proficiency while preserving independence. The patterns demonstrate how AI can effectively address language challenges, create engaging creative spaces, and encourage ongoing, motivated writing, ultimately leading to proficient and emotionally invested EFL writers.

This study builds on existing literature by empirically demonstrating how AI influences intrinsic, extraneous, and germane loads in EFL CW, in line with CLT as outlined by[62,65]. The findings suggest that these tools alleviate mental effort related to technical aspects, allowing learners to focus more on schema-building for creative expression[60,94]. This work deepens the conversation around EE, drawing on the insights of[51]. It demonstrates how AI can foster positive emotions, such as enthusiasm, in language tasks. This aligns with[46] perspective on engagement as a motivational disposition that thrives through non-judgmental interactions, as well as the autonomy and competence needs outlined in SDT[96,104]. Furthermore, it promotes CW in EFL by emphasizing the importance of AI co-creation in fostering originality and playfulness. This builds on the work of[4,5] to illustrate how AI can inspire innovative language use, while also addressing the ethical and pedagogical challenges of integrating AI for learner development[2,3,33,89,99].

The implications offer focused, practical strategies for EFL stakeholders to effectively utilize AI. Learners can effectively incorporate ChatGPT into their writing process by utilizing high support for iterative drafting. For instance, asking, "Refine this sentence for emotional depth in a story about loss," reduces CL on syntax and increases engagement through immediate improvements, as reported by High-AI participants. Alternatively, low support for autonomy-building can be employed by requesting, "List synonyms for resilience," prior to self-revising, which strengthens grammar and enhances self-efficacy according to Low-AI insights. In resource-limited settings, educators can utilize AI for initial essay reviews, allowing for immediate corrections of mechanical errors. This approach frees up class time for collaborative discussions focused on creativity and cultural relevance. Additionally, conducting workshops on prompt engineering can empower students, while monitoring usage through logs helps to mitigate issues of over-reliance or biases[87]. Institutions' policymakers ought to prioritize investments in AI platforms that incorporate privacy-compliant features, such as encrypted data handling. Additionally, they should implement pilot programs in under-resourced areas to promote access equity, while monitoring metrics like engagement scores to enhance policies and address proficiency gaps. Curriculum designers have the opportunity to integrate AI modules into their syllabi. This can involve creating progressive tasks that range from low to high support, effectively scaffolding students' skills from basic conventions to more advanced narratives. Evaluating outcomes can be achieved through pre- and post-assessments.

For those developing educational materials, creating AI-enhanced textbooks requires the inclusion of adaptive exercises with varying levels of complexity. For instance, beginner-level grammar aids can progress to advanced creativity prompts, such as "Generate alternative endings for this plot." It's essential to ensure cultural sensitivity and reduce biases by utilizing diverse datasets during training. Additionally, user interfaces should incorporate progress trackers to facilitate self-monitoring[47]. Administrators have the opportunity to promote collaboration across departments to develop AI ethics guidelines. By incorporating these tools into professional development,

they can strike a balance between technology and humanistic teaching, thereby nurturing inclusive and effective learning environments.

Limitations and suggestions for future research

It is important to recognize certain limitations that were placed on the study. This study faces a significant limitation due to the exclusive participation of male subjects, stemming from Iran's gender-segregated educational policies that required recruitment from male-only settings. This restriction notably limits the applicability of the findings to female EFL learners or co-educational contexts, where gender dynamics could affect responses to AI-supported interventions in distinct ways. This study faced several limitations, including a modest sample size of 92 intermediate male EFL learners from two institutes in Ahvaz, which was selected through convenience sampling. This approach raises concerns about bias and restricts the applicability of findings to female learners, different proficiency levels, or non-Iranian contexts, particularly due to gender-segregated policies and an urban focus. Additionally, the short-term design did not account for long-term retention or skill transfer. The reliance on self-reported measures for EE and CL may introduce subjectivity. Furthermore, the exclusion of advanced AI features like multimodal inputs limited the depth of the analysis, and the potential confounding effects of participants' varying levels of technology familiarity were not fully addressed. To address these challenges, future studies could focus on creating culturally adaptive AI models that utilize NLP to tailor feedback for various demographics. This approach would facilitate multinational trials aimed at comparing the effects of gender and proficiency on engagement. Utilizing longitudinal tracking through wearable biosensors could provide insights into real-time physiological indicators of CL and emotion during prolonged AI engagement, highlighting lasting effects on skill retention, such as in speaking. Investigating the role of hybrid AI-human mentoring within virtual reality presents an opportunity to evaluate multimodal transfers, such as transitioning from writing to oral storytelling, while incorporating randomized controls for technological proficiency. Moreover, incorporating ethical AI audits through blockchain technology for clear data usage can help mitigate subjective biases. At the same time, exploring gamified AI interfaces could boost engagement among underrepresented groups, thereby increasing their relevance and reach.

Data availability

The findings of this study are supported by data that can be obtained from the corresponding author upon reasonable request.

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Author contributions

Goodarz Shakibaei was responsible for the conceptualization, study design, data collection, data analysis, and drafting of the original manuscript. Xinqiao Cen contributed significantly to revising the manuscript based on the reviewers' comments, refining the structure and clarity of the paper, and providing extensive support in proofreading and editing the final version. Both authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

This study was conducted in accordance with the ethical standards of the Declaration of Helsinki. Ethical approval was waived by institutional regulations, as the study involved anonymous, non-invasive educational research with voluntary participation. Written informed consent was obtained from all participants prior to data collection.

AI Use Statement

Generative artificial intelligence tools (ChatGPT and Grammarly) were used exclusively for linguistic and stylistic assistance, including improvinggrammar, clarity, coherence, and overall readability of the manuscript. These tools were not used for data collection, data analysis, interpretation of results,or the generation of scholarly content. All intellectual contributions, academic judgments, theoretical framing, methodological decisions, and interpretationsare entirely the responsibility of the authors.

Additional information

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